VICTORIA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 1

CHEMISTRY 8872/01

Paper 1 Multiple Choice

19 September 2012

50 mins

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your NRIC/FIN number, name and CT group on the Answer Sheet.

There are **thirty** questions. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choices in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

This document consists of **11** printed pages.

Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

1 *Use of the Data Booklet is relevant to this question.*

Oxides of nitrogen are pollutant gases which are emitted from car exhausts. In urban traffic, when a car travels one kilometre, it releases 0.23 g of an oxide of nitrogen N_aO_b, which occupies 120 cm^3 at room temperature and pressure.

What are the values of *a* and *b*?

A $a = 2$, $b = 4$

B $a = 2$, $b = 1$

C $a = 1$, $b = 1$

D $a = 1$, $b = 2$

2 'Black powder' consisted of a mixture of potassium nitrate, sulfur and charcoal (carbon) in the molar ratio shown in the equation below, was widely used as propellant for muzzle-loading cannons.

$$2KNO_3(s) + S(s) + 3C(s) \rightarrow K_2S(s) + N_2(g) + 3CO_2(g)$$

Given that 2.00 mol of gas could just expel the cannonball out of the cannon tube, what is the minimum mass of 'black powder', in g, required for this to happen?
[M_r: KNO_3, 101]

A 101 **B** 135 **C** 202 **D** 270

3 It was believed that the bombardment of polonium atoms, Po, with atoms of an isotope of a Group III element, **X**, may bring about a new element with proton number 115 via the following reaction.

$$^{209}_{84}Po + \mathbf{X} \rightarrow \text{[new element]} + 5 \text{ neutrons}$$

What is the identity of **X**?

A aluminium **B** gallium **C** indium **D** thallium

4 The successive ionisation energies (IE) of two elements, **Q** and **R**, are given below.

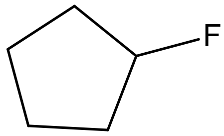
IE / kJ mol^{-1}	1st	2nd	3rd	4th	5th	6th	7th	8th
Q	1090	2350	4610	6220	37800	47000	-	-
R	1251	2298	3822	5158	6542	9362	11018	33604

What is the likely formula of the compound that is formed when **Q** reacts with **R**?

A **QR** **B** $\mathbf{Q_2R_3}$ **C** $\mathbf{QR_4}$ **D** $\mathbf{Q_4R}$

5 The following compounds all have M_r = 88. Which one contains over 60% by mass of carbon **and** also exhibits hydrogen bonding?

A $H_2N(CH_2)_4NH_2$

B 

C $CH_3(CH_2)_4OH$

D $CH_3CH_2CH_2CO_2H$

6 In the presence of ultraviolet light, the “inert” xenon gas will react with fluorine gas to produce XeF_4 according to the equation,

$$Xe(g) + 2F_2(g) \rightleftharpoons XeF_4(s)$$

Which is the correct expression for the equilibrium constant, K_c?

A $\dfrac{[Xe][F_2]}{[XeF_4]}$ **B** $\dfrac{[XeF_4]}{[Xe][F_2]}$ **C** $\dfrac{[XeF_4]}{[Xe][F_2]^2}$ **D** $\dfrac{1}{[Xe][F_2]^2}$

7 $$CH_3CO_2H + C_2H_5OH \rightleftharpoons CH_3CO_2C_2H_5 + H_2O$$

The above reaction can be said to have reached equilibrium when

A the equilibrium constant K is equal to 1.

B the reaction between the acid and the alcohol has stopped.

C the concentrations of the products equal those of the reactants.

D the rate of production of ethyl ethanoate equals its rate of hydrolysis.

8 Two equilibria are shown below.

Reaction I: $\mathbf{X}_2\mathbf{Y}(g) \rightleftharpoons \mathbf{X}_2(g) + \frac{1}{2}\mathbf{Y}_2(s)$
Reaction II: $2\mathbf{X}_2(g) + \mathbf{Y}_2(s) \rightleftharpoons 2\mathbf{X}_2\mathbf{Y}(g)$

The numerical value of K_c for reaction I is 2. Under the same conditions, what is the numerical value of K_c for reaction II?

A $\frac{1}{4}$ **B** 4 **C** $\sqrt{\frac{1}{2}}$ **D** $\frac{1}{2}$

9 A 2.0 dm^3 solution of a strong acid has pH 2. What volume of water, in dm^3, is added to it so as to increase the pH of the solution to pH 3?

A 1.0 **B** 3.0 **C** 18.0 **D** 20.0

10 Which statement about $AlCl_3$ is correct?

A $AlCl_3$ is pyramidal.

B $AlCl_3$ has a higher melting point than Al_2O_3.

C The Al_2Cl_6 dimer contains two co-ordinate bonds.

D An aqueous solution of $AlCl_3$ is neutral.

11 The reaction between substances **A** and **B** is found to follow the rate law

$$\text{rate} = k[\mathbf{A}]^m[\mathbf{B}]$$

where k is the rate constant and has units of $mol^{-2}\ dm^6\ s^{-1}$.

Two experiments to study the kinetics of this reaction were carried out and the data obtained are tabulated below.

Experiment	Initial [**A**] / $mol\ dm^{-3}$	Initial [**B**] / $mol\ dm^{-3}$	Initial rate / $mol\ dm^{-3}\ s^{-1}$
1	0.040	0.080	S
2	0.020	y	$S/2$

What is the value of y in experiment 2?

A 0.020 **B** 0.040 **C** 0.160 **D** 0.320

12 Which is the correct statement about the following reaction?

$$\mathbf{P}(g) + \mathbf{Q}(g) \xrightleftharpoons{\mathbf{M}(s)} \mathbf{R}(g) \qquad \Delta H < 0$$

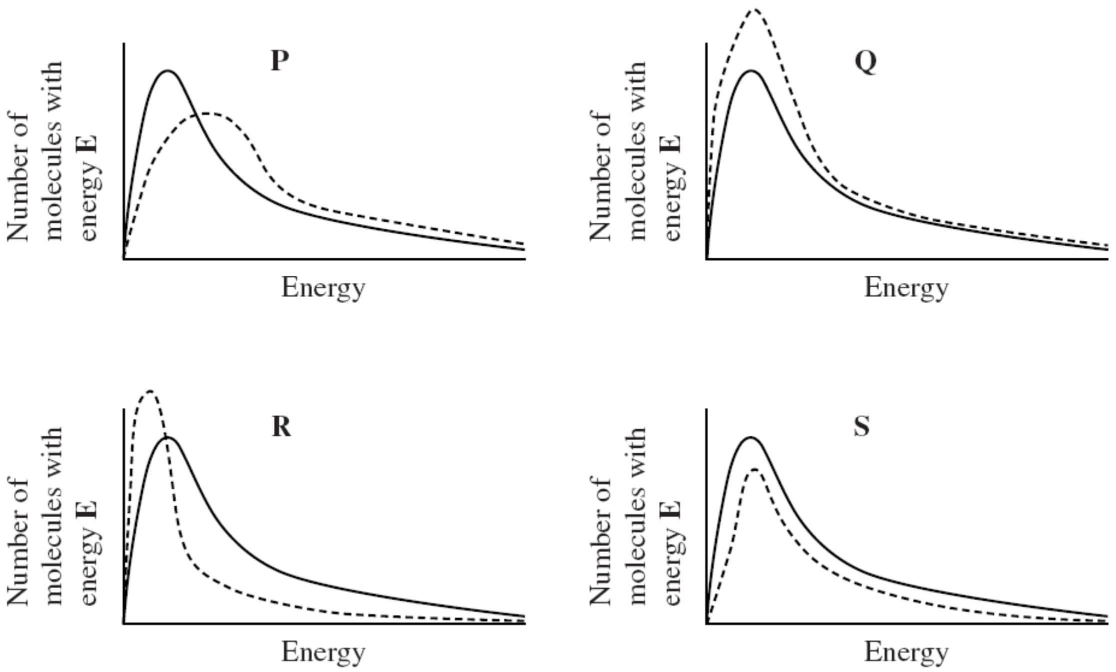
A The solid **M** will lower the activation energy of both forward and backward reactions.

B Both the rate constant and equilibrium constant will increase with increasing temperature.

C Increasing temperature will lower the activation energy resulting only in a greater fraction of **R** molecules with energy greater than activation energy.

D The activation energy of the forward reaction is equal to the activation energy of the backward reaction.

13 The diagrams **P**, **Q**, **R** and **S** show how a change in conditions affects the Maxwell-Boltzmann distribution of molecular energies for gas **G**. In each case, the original distribution is shown by a solid line and the distribution after a change has been made is shown by a dashed line.



Which statement about the change made is correct?

A The change shown in diagram **Q** occurs when a catalyst is used.

B The change shown in diagram **R** occurs when the temperature is increased.

C The change shown in diagram **P** occurs when the temperature is decreased.

D The change shown in diagram **S** occurs when less gas **G** is present at constant temperature.

14 Given the enthalpy change of the following reaction is –310 kJ mol^{-1}.

$$C_2H_2(g) + 2H_2(g) \rightarrow C_2H_6(g)$$

Given that the enthalpy change of combustion of ethyne (C_2H_2) and ethane (C_2H_6) are –1300 kJ mol^{-1} and –1560 kJ mol^{-1} respectively, what is the enthalpy change of combustion of hydrogen?

A +25 kJ mol^{-1}

B +50 kJ mol^{-1}

C –285 kJ mol^{-1}

D –570 kJ mol^{-1}

15 For the elements (Na to S) in the third period, the oxide of element **U** has the highest melting point and the pH of the chloride of element **V** is around 2. Which elements could **U** and **V** be?

	U	V
A	magnesium	silicon
B	aluminium	silicon
C	silicon	aluminium
D	silicon	phosphorus

16 Which of the following statements about Period 3 elements and their compounds is **incorrect**?

A Electrical conductivity of elements decreases in the order A*l* > Mg > Na > Si.

B Melting point of elements decreases in the order P > S > C*l* > Ar.

C Ionic radius of potassium ion is smaller than that of chloride ion.

D Atomic radius of potassium is bigger than that of chlorine.

17 A compound with a general formula $C_nH_{2n}O_2$ does **not** contain

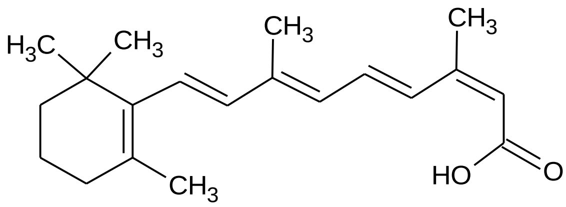
A two carbonyl groups

B one carbonyl and one hydroxyl group

C one carboxylic acid group

D one ester group

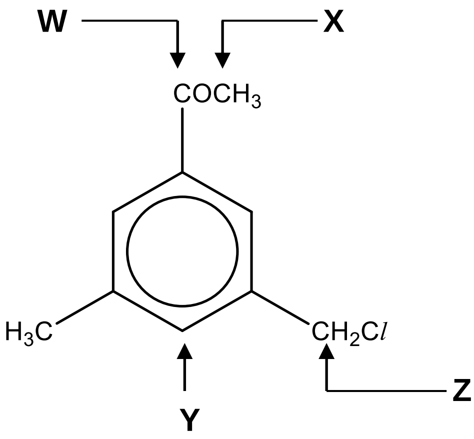
18 *Isotretinoin*, a common drug used in the treatment of severe acne has the following structure.



What is the total number of geometric isomers in Isotretinoin?

A 2^3 **B** 2^4 **C** 2^5 **D** 2^6

19 Which carbon atoms indicated in the molecule below are trigonal planar?



A **X** and **Y** only

B **Y** and **Z** only

C **W** and **X** only

D **W** and **Y** only

20 Compound **Y** undergoes the following reactions:

$$\mathbf{Y} \xrightarrow{\text{ethanolic KOH}} C_4H_8 \xrightarrow{KMnO_4/H_2SO_4} CH_3COCH_3 + CO_2$$

Compound **Y** could be

A $CH_2ICH(CH_3)_2$

B $(CH_3)_2CHCl$

C $CH_3CH_2CHICH_3$

D $CH_3CH_2CH_2CH_2I$

21 Compound **G** is boiled with aqueous sodium hydroxide and the resulting mixture cooled and acidified with dilute sulfuric acid. The final products include a compound $C_3H_6O_2$ and an alcohol. This alcohol gives a positive iodoform test.

What is the formula for compound **G**?

A $CH_3CH_2CO_2CH_3$

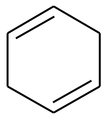
B CH_3CH_2 $OCOCH_3$

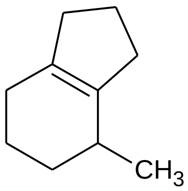
C $CH_3CH_2CO_2CH_2CH_3$

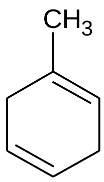
D $HOCH_2CH_2$ $CO_2CH_2CH_3$

22 Which of the following compounds form two organic products when it is heated with acidified potassium manganate(VII) solution?

A $CH_2{=}CHCH_2CH{=}CH_2$

B 

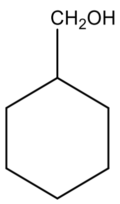
C 

D 

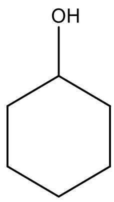
23 On dehydration followed by reaction with chlorine in tetrachloromethane, compound **Z** produced a mixture of two dichloro compounds that are structural isomers.

What could compound **Z** be?

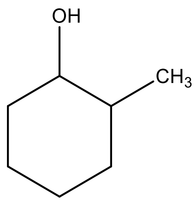
A



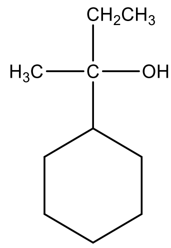
B



C



D



24 An organic compound undergoes the following reactions.

(i) It decolourises a solution of bromine in tetrachloromethane.

(ii) It reacts with phosphorus pentachloride giving copious white fumes of HC*l*.

(iii) It reacts with hot alkali to give a compound with two hydroxyl groups.

Which of the following represents the organic compound?

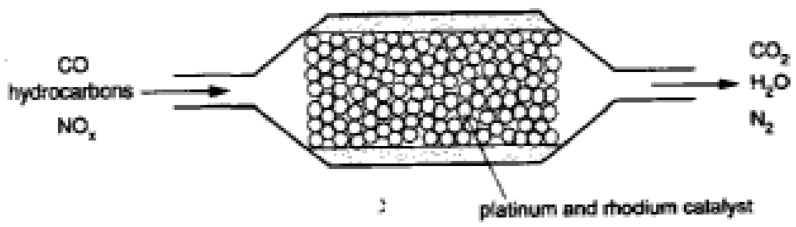
A $ClCH_2CH_2CH{=}CHCH(Cl)CH_2OH$

B $ClCH_2CH_2CH{=}CHCO_2H$

C $BrCH_2CH_2CH_2CH_2CH_2Cl$

D $HOCH_2CH{=}CHCH{=}CHCH_2Cl$

25 The diagram represents a section of a catalytic converter on the exhaust system for a car. Harmful gases are converted into carbon dioxide, nitrogen and water vapour.



Which of the following statements is **incorrect**?

A Carbon monoxide and hydrocarbons react together.

B Carbon monoxide and nitrogen oxide react together.

C Platinum and rhodium catalyse redox reactions.

D Presence of lead will lead to poisoning of platinum and rhodium catalysts.

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct.)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

26 In which of the following compounds is the bond angle in molecule **I** greater than that in molecule **II**?

	I	**II**
1	BCl_3	NCl_3
2	NO_3^-	ClO_2^-
3	SF_6	SF_4^{2-}

27 The enthalpy change of formation of carbon monoxide and carbon dioxide are given below.

$\Delta H_f(CO) = -110\ kJ\ mol^{-1}$
$\Delta H_f(CO_2) = -393\ kJ\ mol^{-1}$

Which of these statements are correct?

1 Carbon monoxide has an exothermic enthalpy change of combustion.

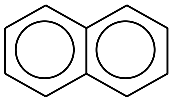
2 Carbon dioxide is energetically more stable than carbon monoxide.

3 The enthalpy change of combustion of carbon is $-393\ kJ\ mol^{-1}$.

28 Which trends across the third period (Na to S) are always correct?

1 The bonding in the oxide changes from ionic to covalent.
2 The first ionisation energy generally increases.
3 The ionic radius decreases.

29 Naphthalene, $C_{10}H_8$, is an aromatic compound with the following structure.



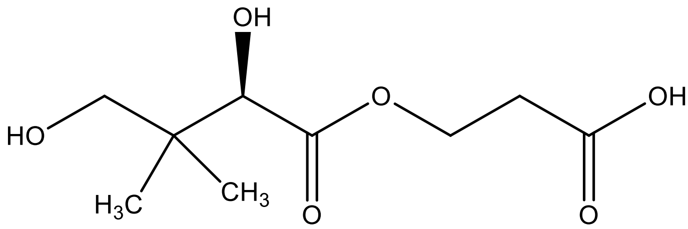
Which of the following statements about naphthalene and its derivatives are correct?

1 Naphthalene will decolourise liquid bromine in the presence of iron(III) chloride.

2 The molecule of naphthalene is planar.

3 There are four aromatic isomers of molecular formula $C_{10}H_7Br$.

30 Vitamin B5 has the following structure.



Which of the following statements are correct?

1 1 mol of Vitamin B5 reacts with 3 mol of PCl_5.

2 1 mol of Vitamin B5 reacts with 1 mol of NaOH at room temperature.

3 Vitamin B5 reacts with 2,4-dinitrophenylhydrazine to give an orange precipitate.